

PHISHING-LED COMPROMISE WITH DATA EXFILTRATION

Date of Detection: Demonstrated by Analyst (Project Demonstration)

Severity Level: CRITICAL

Affected Assets: LAPTOP-UMIT35BS (Compromised Host), WIN-SERVER (Target Server), alex (User Account), 45.227.10.23 (External C2 IP)

EXECUTIVE SUMMARY

A phishing-led attack sequence was successfully executed and analyzed as a project demonstration. The simulation involved a complete multi-stage corporate network compromise, progressing through initial email-based phishing, web-based credential dumping, local discovery, malicious script execution, lateral movement, data exfiltration, and persistent Command-and-Control (C2) communication. Threat detection, lifecycle correlation, and behavioral tracking were fully validated using Splunk SIEM dashboards and event logs.

MITRE ATT&CK MAPPING

- **Phishing:** T1566
- **Credential Dumping:** T1003
- **PowerShell Execution:** T1059.001
- **Remote Services (PsExec):** T1021
- **Exfiltration Over HTTP:** T1041
- **Application Layer Protocol (C2 Communication):** T1071

DETAILED TIMELINE OF EVENTS

TIMESTAMP	EVENT DETAILS & CORRELATED ACTIVITY
15:43:45	User initiated a connection and visited the malicious external URL (http://malicious-login.xyz).
15:44:10	Credential dumping activity identified on the endpoint targeting the active session of user alex.
15:45:02	Execution of an unverified/malicious PowerShell script (evil.ps1) detected locally on the host.

TIMESTAMP	EVENT DETAILS & CORRELATED ACTIVITY
15:45:30	Lateral movement phase: Remote service execution initiated to WIN-SERVER leveraging PsExec.
15:46:55	Data exfiltration event: Sensitive corporate file salary.xlsx successfully uploaded to an attacker-controlled external infrastructure.
15:47:22	Continuous Command-and-Control (C2) beaconing patterns flagged out to malicious IP destination 45.227.10.23.
15:47:40	Unauthorized internal interactive login established on WIN-SERVER utilizing compromised credentials belonging to alex.
15:48:05	Anomalous and suspicious network outbound traffic spikes flagged across network boundaries.

IMPACT ASSESSMENT

- Credential compromise risk
- Unauthorized access to WIN-SERVER
- Sensitive data exfiltration activity
- Persistent external communication risk
- Potential data exposure and security risk

ROOT CAUSE ANALYSIS

Primary Root Cause

Successful phishing activity leading to credential compromise.

Contributing Factors

- User interaction with malicious URL
- PowerShell misuse
- Lack of outbound traffic restriction
- Weak endpoint monitoring

DETECTION METHOD

- Splunk log analysis
- Dashboard monitoring
- Alert correlation
- Security event investigation

ACTIONS PERFORMED

- Uploaded phishing logs into Splunk
- Investigated phishing URL access
- Identified credential dumping activity
- Investigated PowerShell execution
- Detected lateral movement and exfiltration
- Created dashboards and alerts

RECOMMENDED CONTAINMENT ACTIONS

Immediate Actions

- Isolate affected host
- Reset compromised credentials
- Block malicious domains and IP addresses
- Investigate WIN-SERVER for persistence
- Review user and network activity

Long-Term Recommendations

- Deploy EDR monitoring
- Restrict PowerShell execution
- Enable outbound traffic monitoring
- Conduct phishing awareness training
- Improve email and endpoint security controls

CONCLUSION

The investigation confirmed a phishing-led compromise involving credential theft, lateral movement, data exfiltration, and C2 communication. Splunk dashboards and alerts successfully detected the attack lifecycle during this demonstration project.